



2 Memory stored in cortex
Networks across the cortex store memories for things that happened less recently. Different types of memories might be stored in various combinations of regions.

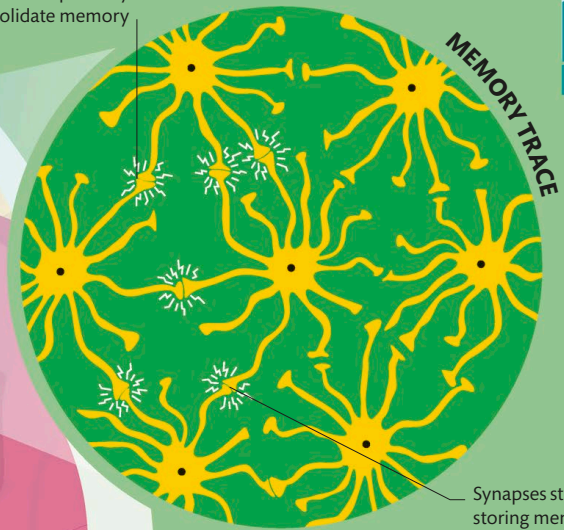
SOMATOSENSORY CORTEX

AUDITORY CORTEX

HIPPOCAMPUS

1 Memory encoded by hippocampus
Experiences are registered by the hippocampus, and some of them—those that are destined to become memories—are encoded there. Long-term potentiation alters connections between neurons in the hippocampus to create a memory. This area is vital for new memories.

A certain combination of neurons fires repeatedly to consolidate memory



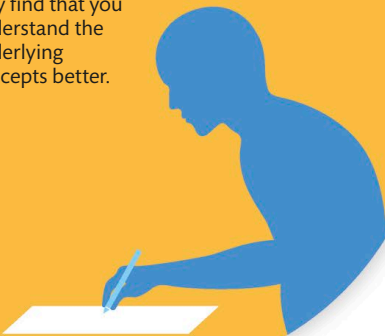
Synapses strengthen, storing memory as a trace

HIPPOCAMPAL INJURY CAN MAKE FORMING NEW LONG-TERM MEMORIES IMPOSSIBLE



RETRIEVAL

3 Remember
When you wake up, the memory of what you learned is stored more securely. It has also been linked to other facts, making it easier to recall, and you may find that you understand the underlying concepts better.



PRACTICE MAKES PERFECT

If you learn something just once, over time that memory trace will fade as the connections weaken. The more times you practice or revise something, the stronger those connections between neurons become and the more likely you are to remember it in the future.

